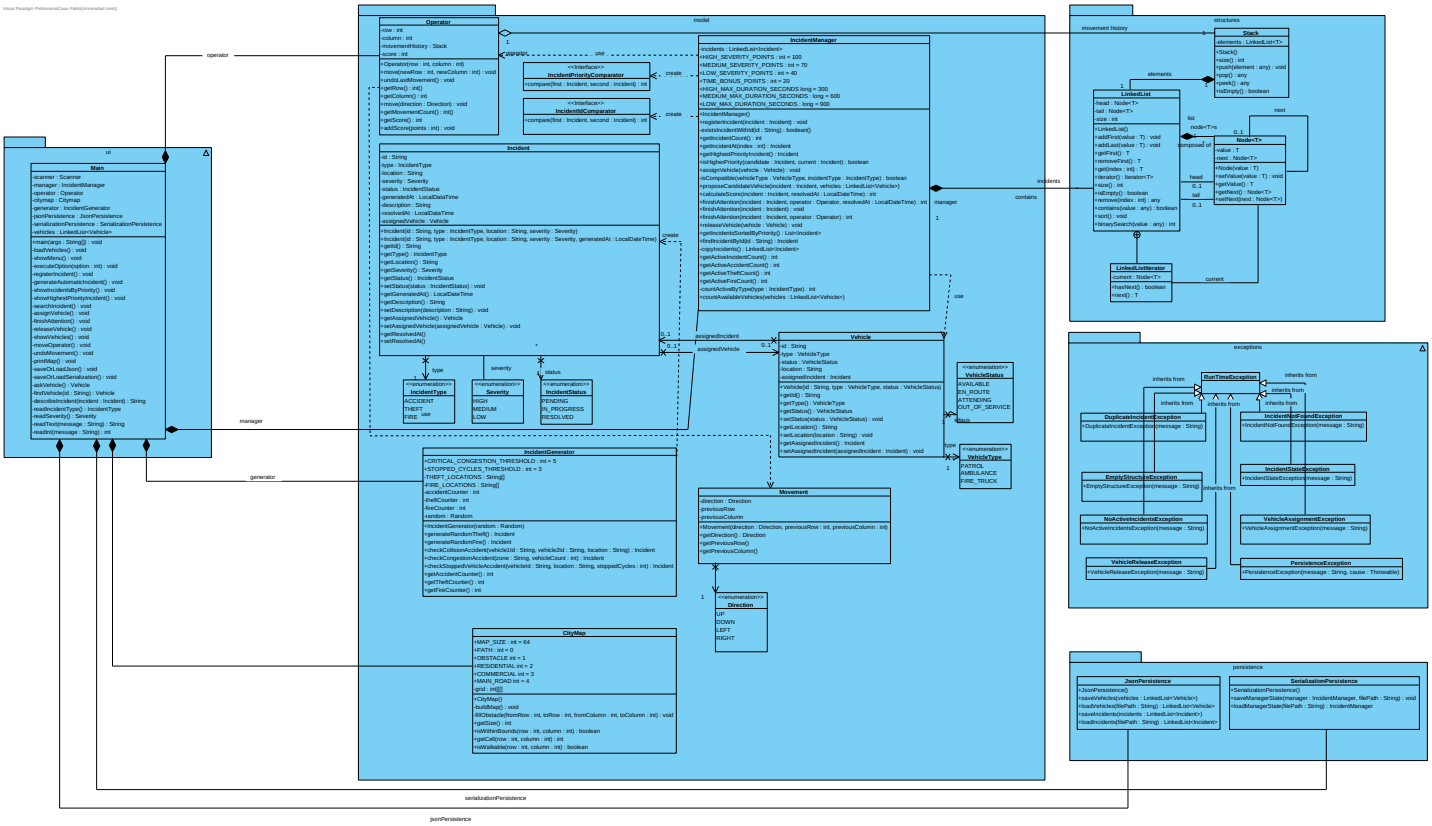
Visual Paradigm Professional(Juan Pablo)(Universidad Icesi)
ui
Main
-scanner : Scanner
-manager : IncidentManager
-operator : Operator
-citymap : Citymap
-generator : IncidentGenerator
-jsonPersistance : JsonPersistance
-serializationPersistence : SerializationPersistence
-vehicles : LinkedList<Vehicle>
+main(args : String[]) : void
-loadVehicles() : void
-showMenu() : void
-executeOption(option : int) : void
-registerIncident() : void
-generateAutomaticIncident() : void
-showIncidentsByPriority() : void
-showHighestPriorityIncident() : void
-searchIncident() : void
-assignVehicle() : void
-finishAttention() : void
-releaseVehicle() : void
-showVehicles() : void
-moveOperator() : void
-undoMovement() : void
-printMap() : void
-saveOrLoadJson() : void
-saveOrLoadSerialization() : void
-askVehicle() : Vehicle
-findVehicle(id : String) : Vehicle
-describeIncident(incident : Incident) : String
-readIncidentType() : IncidentType
-readSeverity() : Severity
-readText(message : String) : String
-readInt(message : String) : int
operator
manager
generator
model
Operator
-row : int
-column : int
-movementHistory : Stack
-score : int
+Operator(row : int, column : int)
+move(newRow : int, newColumn : int) : void
+undoLastMovement() : void
+getRow() : int()
+getColumn() : int
+move(direction : Direction) : void
+getMovementCount() : int()
+getScore() : int
+addScore(points : int) : void
use
<<Interface>>
IncidentPriorityComparator
+compare(first : Incident, second : Incident) : int
create
<<Interface>>
IncidentIdComparator
+compare(first : Incident, second : Incident) : int
create
IncidentManager
-incidents : LinkedList<Incident>
+HIGH_SEVERITY_POINTS : int = 100
+MEDIUM_SEVERITY_POINTS : int = 70
+LOW_SEVERITY_POINTS : int = 40
+TIME_BONUS_POINTS : int = 20
+HIGH_MAX_DURATION_SECONDS:long = 300
+MEDIUM_MAX_DURATION_SECONDS : long = 600
+LOW_MAX_DURATION_SECONDS : long = 900
+IncidentManager()
+registerIncident(incident : Incident) : void
+existsIncidentWithId(id : String) : boolean()
+getIncidentCount() : int
+getIncidentAt(index : int) : Incident
+getHighestPriority(incident() : Incident
+setHighestPriority(candidate : Incident, current : Incident) : boolean
+assignVehicle(vehicle : Vehicle) : void
-isCompatible(vehicleType : VehicleType, incidentType : IncidentType) : boolean
+proposeCandidateVehicle(incident : Incident, vehicles : LinkedList<Vehicle>)
+calculateScore(incident : Incident, resolvedAt : LocalDateTime) : int
+finishAttention(incident : Incident, operator : Operator, resolvedAt : LocalDateTime) : int
+finishAttention(incident : Incident) : void
+finishAttention(incident : Incident, operator : Operator) : int
+releaseVehicle(vehicle : Vehicle) : void
+getIncidentsSortedByPriority() : List<Incident>
+findIncidentById(id : String) : Incident
-copyIncidents() : LinkedList<Incident>
+getActiveIncidentCount() : int
+getActiveAccidentCount() : int
+getActiveTheftCount() : int
+getActiveFireCount() : int
-countActiveByType(type : IncidentType) : int
+countAvailableVehicles(vehicles : LinkedList<Vehicle>)
incidents
contains
manager
1
Incident
-id : String
-type : IncidentType
-location : String
-severity : Severity
-status : IncidentStatus
-generatedAt : LocalDateTime
-description : String
-resolvedAt : LocalDateTime
-assignedVehicle : Vehicle
+Incident(id : String, type : IncidentType, location : String, severity : Severity)
+Incident(id : String, type : IncidentType, location : String, severity : Severity, generatedAt : LocalDateTime)
+getId() : String
+getType() : IncidentType
+getLocation() : String
+getSeverity() : Severity
+getStatus() : IncidentStatus
+setStatus(status : IncidentStatus) : void
+getGeneratedAt() : LocalDateTime
+getDescription() : String
+setDescription(description : String) : void
+getAssignedVehicle() : Vehicle
+setAssignedVehicle(assignedVehicle : Vehicle) : void
+getResolvedAt()
+setResolvedAt()
create
0..1
0..1
assignedIncident
assignedVehicle
0..1
0..1
type
severity
status
<<enumeration>>
IncidentType
ACCIDENT
THEFT
FIRE
<<enumeration>>
Severity
HIGH
MEDIUM
LOW
<<enumeration>>
IncidentStatus
PENDING
IN_PROGRESS
RESOLVED
Vehicle
-id : String
-type : VehicleType
-status : VehicleStatus
-location : String
-assignedIncident : Incident
+Vehicle(id : String, type : VehicleType, status : VehicleStatus)
+getId() : String
+getType() : VehicleType
+getStatus() : VehicleStatus
+setStatus(status : VehicleStatus) : void
+getLocation() : String
+setLocation(location : String) : void
+getAssignedIncident() : Incident
+setAssignedIncident(assignedIncident : Incident) : void
use
1
status
type
1
<<enumeration>>
VehicleStatus
AVAILABLE
EN_ROUTE
ATTENDING
OUT_OF_SERVICE
<<enumeration>>
VehicleType
PATROL
AMBULANCE
FIRE_TRUCK
IncidentGenerator
+CRITICAL_CONGESTION_THRESHOLD : int = 5
+STOPPED_CYCLES_THRESHOLD : int = 3
-THEFT_LOCATIONS : String[]
-FIRE_LOCATIONS : String[]
-accidentCounter : int
-theftCounter : int
-fireCounter : int
-random : Random
+IncidentGenerator(random : Random)
+generateRandomTheft() : Incident
+generateRandomFire() : Incident
+checkCollisionAccident(vehicleId : String, vehicle2Id : String, location : String) : Incident
+checkCongestionAccident(zone : String, vehicleCount : int) : Incident
+checkStoppedVehicleAccident(vehicleId : String, location : String, stoppedCycles : int) : Incident
+getAccidentCounter() : int
+getTheftCounter() : int
+getFireCounter() : int
Movement
-direction : Direction
-previousRow
-previousColumn
+Movement(direction : Direction, previousRow : int, previousColumn : int)
+getDirection() : Direction
+getPreviousRow()
+getPreviousColumn()
1
<<enumeration>>
Direction
UP
DOWN
LEFT
RIGHT
CityMap
+MAP_SIZE : int = 64
+PATH : int = 0
+OBSTACLE int = 1
+RESIDENTIAL int = 2
+COMMERCIAL int = 3
+MAIN_ROAD int = 4
-grid : int[][]
+CityMap()
-buildMap() : void
-fillObstacle(fromRow : int, toRow : int, fromColumn : int, toColumn : int) : void
+getSize() : int
+isWithinBounds(row : int, column : int) : boolean
+getCell(row : int, column : int) : int
+isWalkable(row : int, column : int) : boolean
structures
movement history
1
Stack
-elements : LinkedList<T>
+Stack()
+size() : int
+push(element : any) : void
+pop() : any
+peek() : any
+isEmpty() : boolean
elements
1
1
LinkedList
-head : Node<T>
-tail : Node<T>
-size : int
+LinkedList()
+addFirst(value : T) : void
+addLast(value : T) : void
+getFirst() : T
+removeFirst() : T
+get(index : int) : T
+iterator() : Iterator<T>
+size() : int
+isEmpty() : boolean
+remove(index : int) : any
+contains(value : any) : boolean
+sort() : void
+binarySearch(value : any) : int
list
composed of
node<T>s
0..1
next
Node<T>
-value : T
-next : Node<T>
+Node(value : T)
+setValue(value : T) : void
+getValue() : T
+getNext() : Node<T>
+setNext(next : Node<T>)
head
0..1
tail
0..1
LinkedListIterator
-current : Node<T>
+hasNext() : boolean
+next() : T
current
exceptions
RunTimeException
inherits from
DuplicateIncidentException
+DuplicateIncidentException(message : String)
IncidentNotFoundException
+IncidentNotFoundException(message : String)
EmptyStructureException
+EmptyStructureException(message : String)
IncidentStateException
+IncidentStateException(message : String)
NoActiveIncidentsException
+NoActiveIncidentsException(message : String)
VehicleAssignmentException
+VehicleAssignmentException(message : String)
VehicleReleaseException
+VehicleReleaseException(message : String)
PersistenceException
+PersistenceException(message : String, cause : Throwable)
persistence
JsonPersistence
+JsonPersistence()
+saveVehicles(vehicles : LinkedList<Vehicle>)
+loadVehicles(filePath : String) : LinkedList<Vehicle>
+saveIncidents(incidents : LinkedList<Incident>)
+loadIncidents(filePath : String) : LinkedList<Incident>
SerializationPersistence
+SerializationPersistence()
+saveManagerState(manager : IncidentManager, filePath : String) : void
+loadManagerState(filePath : String) : IncidentManager
serializationPersistence
jsonPersistence